

Control Flow in Python Report

Intern Name: _(Your Name)_

1. Objective

In this task I learned how to use conditional statements and loops in Python. I made two programs -- one that checks numbers and one that does loop operations.

2. What I Did

number_checks.py

This program asks the user to enter a number and then:

- Checks if the number is even or odd
- Checks if the number is positive, negative, or zero

I used if, elif, and else for this. I also used try/except so the program doesn't crash if someone types a word instead of a number.

loop_operations.py

This program has three things:

- Prints a multiplication table for any number the user enters
- Calculates the sum of first N natural numbers
- Prints a star pattern

I used for loop and range() for all three of these.

3. How the Code Works

Even or Odd check:

```
print("Even")
else:
    print("Odd")
```

if number

The % operator gives the remainder. If the remainder is 0 the number is even.

Positive, Negative, Zero check:

```
print("Positive")
elif number < 0:
    print("Negative")
else:
    print("Zero")
```

if number

Multiplication table:

```
print(f"{n} x {i} = {n * i}")
```

```
for i in r
```

Sum of N numbers:

```
for i in range(1, n + 1):  
    total += i
```

```
total = 0
```

Star pattern:

```
print("* " * i)
```

```
for i in r
```

4. Test Cases

Input	Output
4	Even, Positive
7	Odd, Positive
-5	Odd, Negative
0	Even, Zero
abc	Shows error message

Screenshots of all outputs are saved in docs/screenshots/.

5. What I Learned

- How to use if, elif, else in Python
- How for loop works with range()
- How to take input from the user
- How to handle wrong input using try/except
- How to print patterns using loops

6. Conclusion

This was a good task for understanding the basics of Python. Conditional statements and loops are used in almost every program so it was helpful to practice them with simple examples.